

MEMORANDUM

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Project: Safer Streets Priority Finder – Phase 2

Re: Task 4: Model Calibration Approaches

This task aims to calibrate the Bayesian models by identifying a set of parameters that perform well for urban, suburban, and rural locations. This memorandum documents the overall approach, and findings from the calibration process.

The Bayesian Models

The project team developed the Bayesian Model in Task 3. The model formation is presented below.

For each segment the adjusted crashes per year are expressed as:

$$N_{adjusted} = w \times N_{est} + (1 - w) \times N_{obs}$$

With

$$w = \frac{\alpha}{(1 + CV \cdot N_{est})^\beta}$$

Where,

N_{est}, N_{obs} is the estimated and observed crashes per year;

α is the maximum value of the weighting factor that will be assigned to N_{est} , (default = 0.8);

CV is the coefficient of variation of the N_{est} , indicating the confidence in the prior crash estimates;

β is the decay factor that controls how quickly the weight for N_{est} will reduce with increase CV and N_{est} , (default = 1).

The two parameters that are being calibrated in Task 4 are:

- α – max_prior_weight, Maximum weight assigned to the prior crash estimates relative to local observed crash data
- β – decay, Controls how quickly the influence of older crash events diminishes over time in the Bayesian update

Calibration Approach

Case Studies

The project team identified three case studies for model calibration and validation. These case studies capture the urban, suburban, and rural built environment context, and are selected based on national geographic representation and data availability.

- Urban, City of New Orleans, LA (2014–2023; ~181,000 crashes)
- Suburban, City of Albany, OR (2014–2023; ~5,800 crashes)
- Rural, Knox County, OH (2015–2024; ~11,300 crashes)

For each location, crash data were analyzed across three travel modes: pedestrian, bicycle, and motor vehicle.

Calibration and Holdout Approach

Each 10-year dataset was split into two 5-year halves. The primary split used the first 5 years as the **calibration set** and the last 5 years as the **holdout set**. This reflects a realistic deployment scenario in which the model is used on historical data to predict future crash risks.

Evaluation Metrics

Model performance was assessed using two ranking-based metrics, each computed at the top 10% and top 25% of segments:

- **NDCG@k** (Normalized Discounted Cumulative Gain at top k%): Measures whether the highest-crash segments in the holdout set are ranked near the top by the calibration-period score. A perfect ranking yields a score of 1.0; a random ranking approaches 0. This metric penalizes incorrect ranking more heavily at the top of the list, making it well suited to safety prioritization where identifying the higher risk segments matters more.
- **Percentile Capture @k** (Cap@k): Measures the fraction of total holdout-period crash exposure captured within the top k% of segments as ranked by the calibration-period score. Higher values indicate that the model successfully concentrates crash risk in its highest-priority segments.

Baseline Definition

The baseline serves as a benchmark method against which the calibrated model's performance is evaluated. It represents the common practice of using currently observed crash history to prioritize treatment locations, under the assumption that crash trends remain stable over time. Specifically, the first 5-year crash score is used directly to rank segments without any model adjustment. Using actual crash data from the subsequent five years as the holdout set, the quality of this baseline ranking can be measured. Using the baseline, we can evaluate the relative improvement that the model can provide.

Grid Search and Parameter Tuning

A grid search was conducted over the model parameters to evaluate the performance across all metrics, modes, and study areas.

In addition to Bayesian parameters, another parameter “blur_size” is also calibrated. This parameter controls a crash score distribution step applied independently from the Bayesian update. This step distributes each crash's influence on nearby roadway window segments using a linearly decay weighting scheme, where segments closer to the crash center receive greater weight. The purpose is twofold: to account for spatial inaccuracies in the recorded crash location, and to represent crash risk over a longer segment of roadway rather than only the immediate location where a crash occurred. Tuning of this parameter is to evaluate how far each crash should influence nearby segments.

Crashes are joined to a roadway network segmented into 0.05-mile window segments, and “blur_size” specifies how many neighboring segments on each side of the crash's host segment receive a share of the crash's influence. For example, A “blur_size” of 0 applies no smoothing (a single 0.05-mile segment), a “blur_size” of 1 distributes each crash across the host segment plus its two immediate neighbors (three segments, 0.15 mile), and “blur_size” of 2 reaches two neighbors on each side (five segments, 0.25 mile).

The grid search sampled the following values:

Parameter	Values tested
max_prior_weight (α)	0.3, 0.4, 0.5, 0.6, 0.7, 0.8
decay (β)	1, 2, 3, 4, 5, 6
blur_size	0, 1, 2, 3, 4, 6, 8

Results

Selected Parameters

Based on the grid search results and balancing predictive performance with spatial resolution considerations, the following parameters were selected:

- **Decay = 3**, Decay has relatively little effect on model performance across the range tested; a moderate value was chosen.
- **Max Prior Weight = 0.5**, Performance is not sensitive at lower values, but performance of the mode degrades when prior estimates are over-weighted with excessive high value.
- **Blur Size = 4**, Selected to balance predictive performance with spatial resolution (see Discussion for further detail).

Rural, Knox County, OH

The Bayesian model delivers its largest gains in the rural setting, with improvement concentrated in the percentile capture metrics and driven primarily by pedestrian and bicycle modes. The smaller motor vehicle improvement reflects higher motor vehicle crash counts per segment, which pull the Bayesian weighting toward observed data for that mode.

Table 1. Average Performance Across Modes - Rural

Metric	Bayesian Model	Baseline	Δ
Cap@10%	0.564	0.399	+0.165
NDCG@25%	0.545	0.457	+0.088
Cap@25%	0.773	0.521	+0.253
Cap@10%	0.564	0.399	+0.165

Table 2. Performance by Mode - Rural

Mode	Metric	Bayesian Model	Baseline	Δ
Pedestrian	NDCG@10%	0.295	0.224	+0.070
	NDCG@25%	0.370	0.262	+0.108
	Cap@10%	0.449	0.258	+0.191
	Cap@25%	0.684	0.338	+0.346
Bicycle	NDCG@10%	0.500	0.406	+0.094
	NDCG@25%	0.547	0.428	+0.119
	Cap@10%	0.710	0.434	+0.276
	Cap@25%	0.895	0.529	+0.366
Motor Vehicle	NDCG@10%	0.659	0.626	+0.033
	NDCG@25%	0.719	0.680	+0.039
	Cap@10%	0.533	0.504	+0.029
	Cap@25%	0.741	0.695	+0.046

Suburban, Albany. OR

In the suburban setting the Bayesian model provides moderate improvement, again concentrated in Cap@25% for pedestrian and bicycle modes. Motor vehicle performance shows little change, consistent with higher motor vehicle crash counts pulling the Bayesian weighting toward observed data and narrowing the gap with the baseline.

Table 3. Average Performance Across Modes - Suburban

Metric	Bayesian Model	Baseline	Δ
NDCG@10%	0.621	0.606	+0.015
Cap@10%	0.593	0.567	+0.026
NDCG@25%	0.715	0.632	+0.084
Cap@25%	0.851	0.694	+0.157

Table 4. Performance by Mode - Suburban

Mode	Metric	Bayesian Model	Baseline	Δ
Pedestrian	NDCG@10%	0.483	0.460	+0.023
	NDCG@25%	0.626	0.496	+0.130
	Cap@10%	0.580	0.539	+0.041
	Cap@25%	0.877	0.598	+0.279
Bicycle	NDCG@10%	0.442	0.419	+0.023
	NDCG@25%	0.588	0.463	+0.125
	Cap@10%	0.494	0.459	+0.035
	Cap@25%	0.785	0.590	+0.195
Motor Vehicle	NDCG@10%	0.938	0.940	-0.002
	NDCG@25%	0.933	0.936	-0.003
	Cap@10%	0.705	0.704	+0.001
	Cap@25%	0.892	0.893	-0.001

Urban, City of New Orleans, LA

In the urban setting, where crash counts are high across all modes, the Bayesian weighting shifts strongly toward observed data and the adjusted ranking closely tracks the baseline. Gains are largest for bicycle Cap@25% but remain small in absolute terms.

Table 5. Average Performance Across Modes - Urban

Metric	Bayesian Model	Baseline	Δ
NDCG@10%	0.723	0.715	+0.008
Cap@10%	0.545	0.536	+0.009
NDCG@25%	0.782	0.771	+0.011
Cap@25%	0.779	0.764	+0.015

Table 6. Performance by Mode - Urban

Mode	Metric	Bayesian Model	Baseline	Δ
Pedestrian	NDCG@10%	0.665	0.650	+0.015
	NDCG@25%	0.717	0.706	+0.011
	Cap@10%	0.531	0.516	+0.015
	Cap@25%	0.754	0.742	+0.013
Bicycle	NDCG@10%	0.591	0.582	+0.010
	NDCG@25%	0.697	0.674	+0.023
	Cap@10%	0.545	0.534	+0.011
	Cap@25%	0.778	0.747	+0.031
Motor Vehicle	NDCG@10%	0.911	0.912	-0.001
	NDCG@25%	0.933	0.933	0.000
	Cap@10%	0.559	0.559	0.000
	Cap@25%	0.805	0.804	+0.001

Overall, the model adds the most value where it is most needed, in lower-crash-density contexts and for vulnerable road user modes where the observed crash record alone is an unreliable predictor of future risk.

Discussion:

Alternative Split Strategies

In addition to the primary split (first 5 years calibration, last 5 years holdout), two alternative strategies were evaluated. One reverses the temporal order, using the later period for calibration and the earlier period as the holdout. The other uses alternating years for calibration and holdout. Results were broadly consistent across all

three strategies. The primary split remains the preferred scenario because it most closely mirrors real-world deployment conditions.

Parameter Sensitivity

Decay. Across the range tested, decay has relatively little influence on composite model performance. A moderate value of 3 was selected without significant performance tradeoffs.

Max Prior Weight. Model performance is largely insensitive to this parameter at lower values. However, at higher values, over-weighting the background rate at the expense of local crash history begins to degrade performance. A value of 0.5 was selected as a practical upper bound that avoids this degradation.

Blur Size (Crash Score Distribution). Increasing blur size generally improves both NDCG and capture metrics for both model types, because spatial smoothing mitigates noise from year-to-year crash randomness. However, larger blur values reduce **spatial resolution**, as crash risk credit is spread to adjacent segments, reducing the precision of site-specific identification. Performance gains exhibit **diminishing returns** at larger blur sizes, and an "elbow" approach was used to identify blur = 4 as the balance point between improved ranking accuracy and acceptable spatial precision.

SROCC

The Spearman Rank Order Correlation Coefficient (SROCC) was also computed as an additional performance metric. SROCC measures the monotonic association between the model-predicted and holdout-observed rank orderings across all segments, rather than focusing on the top of the distribution. However, SROCC weights each segment equally regardless of its crash risk level, which does not align well with our use case where correctly ranking higher-risk segments is more important than the overall rank ordering. For this reason, NDCG and percentile capture, which focus on the top of the distribution, were selected as the primary evaluation metrics for this analysis.